(1) Find the lengths of the major and minor axes, co-ordinates of vertices, eccentricity, co-ordinates of the foci, equations of directrices and the length of the Latus Rectum of the following conics.

(i) $9x^2+16y^2=144$ (ii) $4x^2+25y^2=100$

(iii) $\frac{x^2}{25}-\frac{y^2}{9}=1$ (iv) $\frac{y^2}{4}-\frac{x^2}{9}=1$

Solution :

(i) Given equation of ellipse is $9x^2+16y^2=144$.

i.e. $\frac{x^2}{16}+\frac{y^2}{9}=1$

$\therefore a^2=16$ and $b^2=9$. $\Rightarrow$ $a=4$ *and* $b=3$

Here $a>b$.

Lengths of axes :-

Length of major axis $=2a=2\times4=8$ units

Length of minor axis is $=2b=2\times3=6$ units

Co-ordinates of vertices:-

Co-ordinates of vertices are $(\pm a,0)$ *and* $(0,\pm b)$

i.e. $(\pm4,0)$ *and* $(0,\pm3)$

Eccentricity :-

$$e=\frac{\sqrt{a^2-b^2}}{a}=\frac{\sqrt{16-9}}{4}=\frac{\sqrt{7}}{4}$$

Co-ordinates of the foci :-

Co-ordinates of the foci $=(\pm ae,0)=\left(\pm4\times\frac{\sqrt{7}}{4},0\right)=(\pm\sqrt{7},0)$.

Equations of Directrices :-

Equation of directrices are $x=\pm\frac{a}{e}$

i.e. $x=\pm\frac{4}{\sqrt{7}/4}=\pm\frac{16}{\sqrt{7}}$

(ii) $4x^2+25y^2=100$

i.e. $\frac{x^2}{25}+\frac{y^2}{4}=1$

$\therefore a^2=25$ and $b^2=4 \Rightarrow a=5$ and $b=2$

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Here $a > b$

Length of axes :

l(major axes) $= 2a = 10$ units

l(minor axis) $= 2b = 4$ units

co-ordinate of vertices:

$(\pm a, 0)$ and $(0, \pm b)$

$(\pm 5, 0)$ and $(0, \pm 2)$

Eccentricity :

$$e = \frac{\sqrt{a^2 - b^2}}{a}$$

$$e = \frac{\sqrt{25-4}}{5} = \frac{\sqrt{21}}{5}$$

Co-ordinates of foci

$(\pm ae, 0) \equiv (\pm \sqrt{2}, 0)$

Equation of Diretrical

$$x = \pm \frac{a}{e}$$

$$x = \pm \frac{5}{\sqrt{21}/5}$$

$$x = \pm \frac{25}{\sqrt{21}}$$

(iii) Given equation of hyperbola is $\frac{x^2}{25} - \frac{y^2}{9} = 1$

$\therefore a^2 = 25$ and $b^2 = 9$. $\Rightarrow$ $a = 5$ *and* $b = 3$

Here $a > b$.

Lengths of axes :-

Length of transverse axis $= 2a = 2 \times 5 = 10$ units

Length of conjugate axis is $= 2b = 2 \times 3 = 6$ units

Co-ordinates of vertices:-

Co-ordinates of vertices are $(\pm a, 0)$,

i.e. $(\pm 5, 0)$

Eccentricity :-

$$e = \frac{\sqrt{a^2 + b^2}}{a} = \frac{\sqrt{25+9}}{5} = \frac{\sqrt{34}}{5}$$

Co-ordinates of the foci :-

Co-ordinates of the foci $= (\pm ae, 0) = \left(\pm 5 \times \frac{\sqrt{34}}{5}, 0\right) = (\pm\sqrt{34}, 0)$.

Equations of Directrices :-

Equation of directrices are $x = \pm \frac{a}{e}$

i.e. $x = \pm \frac{5}{\sqrt{34}/5} = \pm \frac{25}{\sqrt{34}}$

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(iv) $\frac{y^2}{4}-\frac{x^2}{9}=1$

$b^2=4 \quad b^2=9 \Rightarrow b=2, a=3$

Here $b<a$

Length of axes:

l (transverse axis) $=2b=4$

l (conjugate axis) $=2a=6$

Co-ordinates of vertices

$(0,\pm b)\equiv(0,\pm 2)$

Eccentricity

$$e=\frac{\sqrt{a^2+b^2}}{b}=\frac{\sqrt{13}}{2}$$

Co-ordinates of foci

$$(a,\pm be)\equiv(0,\pm\sqrt{13})$$

Equation of directrices

$$y=\pm\frac{b}{e}=\pm\frac{2}{\sqrt{13}/2}$$

$$y=\pm\frac{4}{\sqrt{13}}$$

2) Find the equation of ellipse referred to its principal axes with eccentricity $\frac{3}{4}$ and passing through the point $(6,4)$.

Solution :

Given : $e=\frac{3}{4}$, $P(x_1,y_1)=(6,4)$.

Let equation of ellipse is $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$

As it passes through the point $P=(6,4)$ we get

$\frac{36}{a^2}+\frac{16}{b^2}=1$ - (1)

Also $b^2=a^2(1-e^2)$

$\therefore\ b^2=a^2\left(1-\frac{9}{16}\right)$

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Foci $=(0,\pm be)$ and Vertices are $=(0,\pm b)$

$\therefore (0,\pm be)=(0,\pm 3)$ and $(0,\pm b)=(0,\pm 2)$

Hence, $be=3$ and $b=2$ thus $b^2=4$.

$\therefore e=\frac{3}{2}$

We have relation between a,b and e is $a^2=(be)^2-b^2$

$\therefore a^2=(3)^2-(2)^2=9-4=5$

Thus equation of hyperbola is $\frac{y^2}{4}-\frac{x^2}{5}=1$.

(iii) directrix is $2x+y=1$, focus at $(1,2)$ and eccentricity $\sqrt{3}$.

$e=\sqrt{3}$

Let $P(x,y)$ be any point on hyperbola

Let $S\equiv(1,2)$

$\therefore SP=\sqrt{(x-1)^2+(y-2)^2}$

$SP^2=(x-1)^2+(y-3)^2=x^2-2x+1+y^2-6y+9$

Let $PM\perp$ directrix $2x+y=1$

$\therefore PM=\left|\frac{2x+y-1}{\sqrt{4+1}}\right|$

$PM=\left|\frac{2x+y-1}{\sqrt{5}}\right|$

$PM^2=\frac{(2x+y-1)^2}{5}$

$PM^2=\frac{4x^2+y^2+1+4xy-2y-4x}{5}$

we have $SP=e\,PM$ (Focus - directrix property)

$x^2-2x+1+y^2-6y+3=3\left(\frac{4x^2+y^2+1+4xy-2y-4x}{5}\right)$

$\Rightarrow 7x^2-2y^2+12xy-2x+14y-22=0$

5) An intersecting property of rectangular hyperbola.

Solution : Equation of rectangular hyperbola is $xy=k$ (k is non zero constant) ...(I)

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Tangent is drawn to the curve at point on it whose abscissa is 'a'. Therefore point of contact is $(a, \frac{k}{a})$

Slope of tangent $=\left[\frac{d}{dx}\left(\frac{k}{x}\right)\right]$ at point $(a, \frac{k}{a})$

$$=\left(\frac{-k}{x^2}\right) \text{ at point } (a, \frac{k}{a})$$

$$=-\frac{k}{a^2} \quad \ldots\ldots\ldots\ldots\ldots\ldots (II)$$

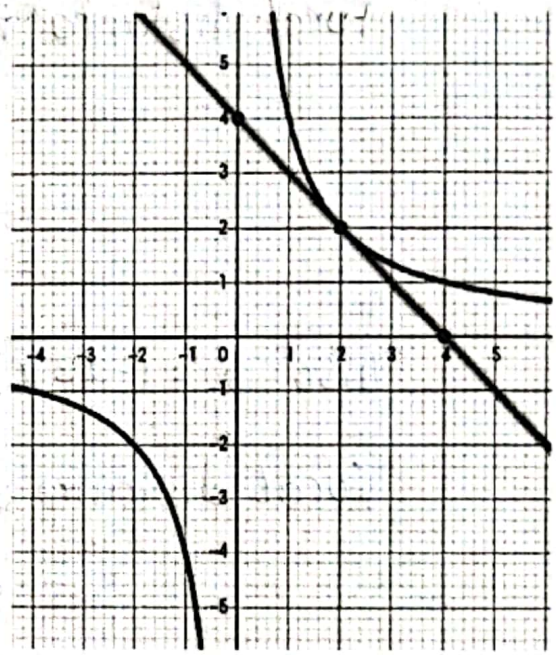
Equation of tangent to $xy=k$ at point $(a, \frac{k}{a})$ is (by slope point form)

$$y-\frac{k}{a}=\left(-\frac{k}{a^2}\right)(x-a)$$

i.e. $a^2\left(y-\frac{k}{a}\right)=-k(x-a)$

i.e. $a^2y-ak=-kx+ak$

i.e. $kx+a^2y-2ak=a$



This equation of tangent in terms of double intercept form is $\frac{x}{2a}+\frac{y}{\frac{2k}{a}}=1.$

Tangent cuts the X – axis at point P and the Y – axis at point Q.

Clearly P = $(2a, 0)$ and Q = $(0, \frac{2k}{a})$

Area of $\Delta POQ=\frac{1}{2}(OP)(OQ)$

$$=\frac{1}{2}(2a)\left(\frac{2k}{a}\right) \quad =2k$$

6) Find the focal distance of the point $P(5,4\sqrt{3})$ on the ellipse $16x^2+25y^2=1600.$

Solution : Given ellipse : $16x^2+25y^2=1600$

Dixiding b.s by 1600

$$\frac{x^2}{100}+\frac{y^2}{64}=1$$

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$a^2 = 100, b^2 = 64$

$e = \frac{\sqrt{a^2-b^2}}{a} = \frac{\sqrt{100-64}}{10} = \frac{\sqrt{36}}{10} = \frac{6}{10} = \frac{3}{5}$

$\therefore ae = 10 \times \frac{3}{5} = 6$

Focus in $S \equiv (ae, 0) \equiv (6, 0)$ and $P(5, 4\sqrt{3})$

Focal distance $PS = \sqrt{(6-5)^2 + (4\sqrt{3}-0)^2}$

$PS = \sqrt{(1)^2 + (4\sqrt{3})^2}$

$PS = \sqrt{1+48} = \sqrt{49}$

$\boxed{PS = 7}$

Also $S' = (-ae, 0) \equiv (-6, 0)$

Focal distance $S'P = \sqrt{(-6-5)^2 + (0-4\sqrt{3})^2}$

$S'P = \sqrt{121+48}$

$S'P = \sqrt{169}$

$\boxed{S'P = 13}$

Focal distanes are 7 and 13

Sign of Teacher :

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